Computer Networks

Minor Project2

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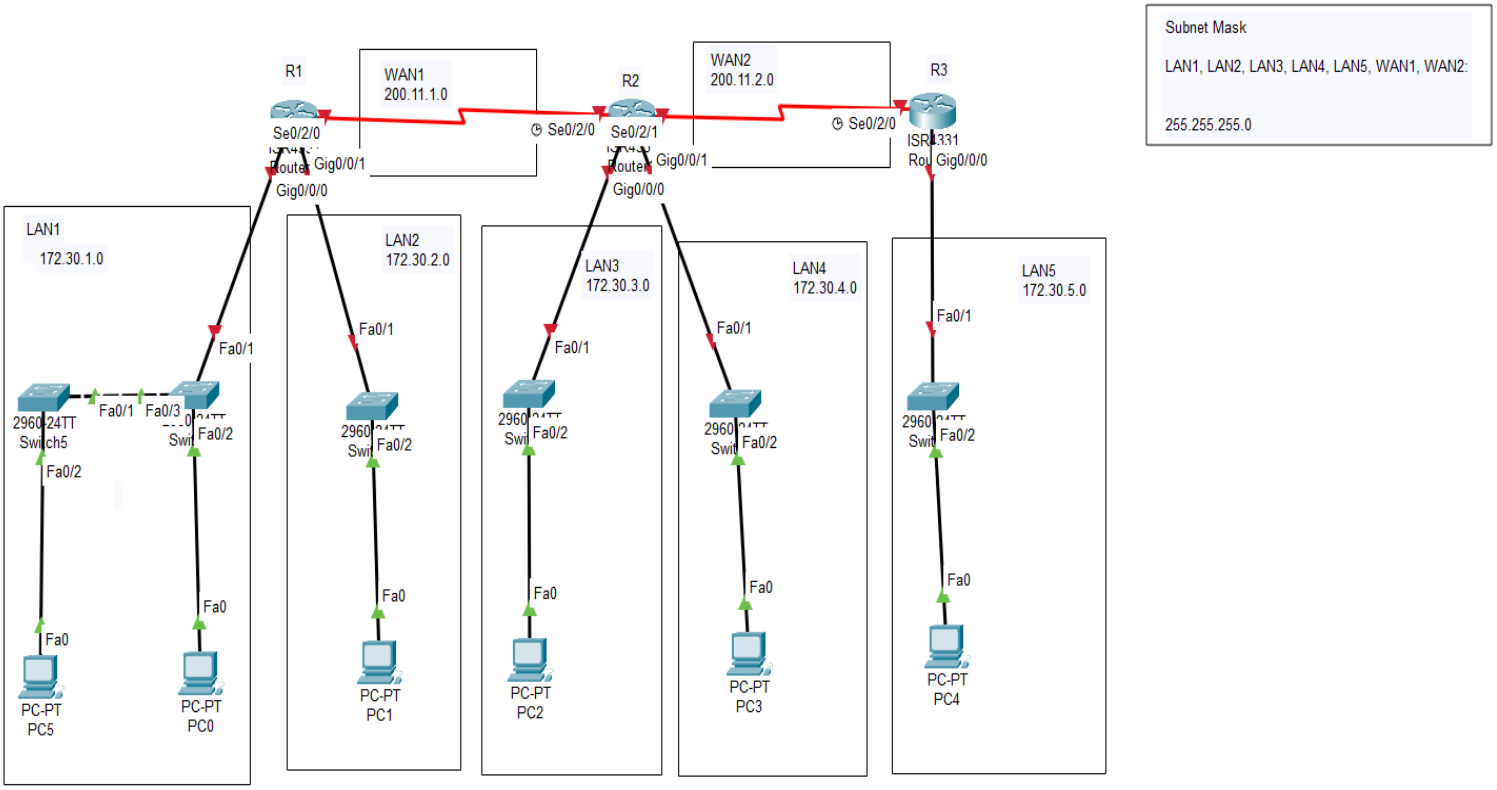
Program البرنامج	Course Title اسم المقرر	Course Code رمز المقرر	Level المستوى	No. of Pages عدد الصفحات	No. of questions عدد الأسئلة
بكالريوس	شبكات الحاسب	503442-3	8	1	5

Question السؤال	Question A السؤال أ	Question B السؤال ب	Question C السؤال ج	Question D السؤال د	Question E السؤال ه	Total المجموع
Designated Marks الدرجة المقررة						
Actual Marks الدرجة الفعلية	1	1	1	1	1	5
Score in words الدرجة كتابة					Numerical score الدرجة بالارقام	
Instructor Name اسم المصحح	Dr. Basma Oueslati				Signature التوقيع	

ُمستعينا بالله، يرجي الاجابة علي جميع الأسئلة في الصفحات التاليةُ

مع تمنياتنا لكم بالتوفيق والنجاح

Network Project



1. Use packet tracer to create the network topology.
2. Configuring device by Adding the IP configuration for each device (PC, Router)
3. Test the ping command between devices in the same Network.
4. Add the routing protocol RIP
5. Test the ping command between device in different Network.

Note:

1. You must take a snapshot for each step and add it in the report.
2. You must send the packet tracer file and the report.

Good luck

1. Use packet tracer to create the network topology.

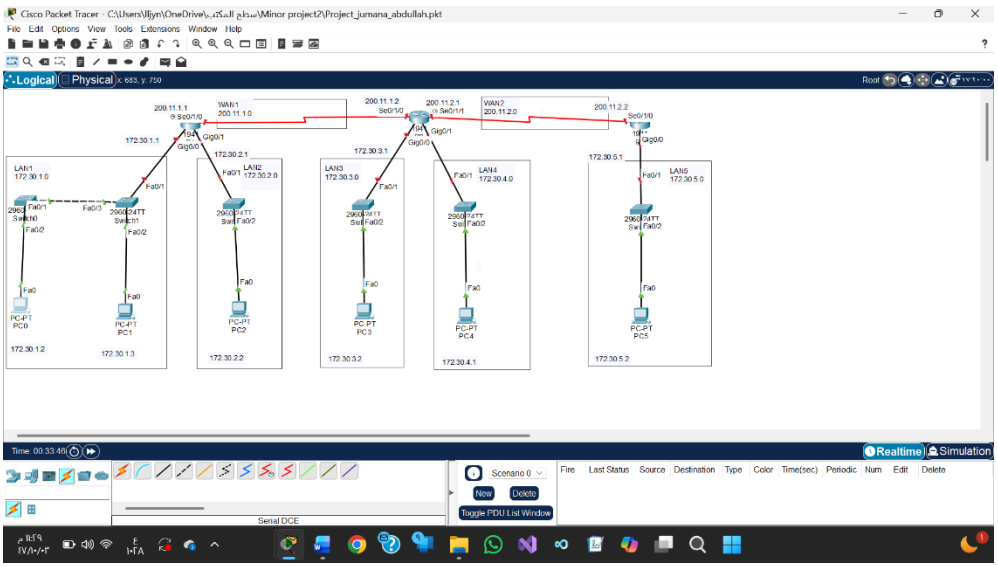


Figure 1.1: Complete Network Topology Design in Cisco Packet Tracer.

The network architecture consists of 3 routers (R1, R2, R3) interconnecting 5 Local Area Networks (LANs) and 2 Wide Area Networks (WANs) and All devices are configured using a standard Subnet Mask of 255.255.255.0 to ensure consistent addressing across the topology.

2. Configuring device by Adding the IP configuration for each device (PC, Router)

In this step, static IPv4 addresses were assigned to all end devices (PCs) and router interfaces. For each PC, the Default Gateway was set to the corresponding router interface IP to enable communication outside its local network. All interfaces were activated (No Shutdown) to establish physical and logical connectivity.

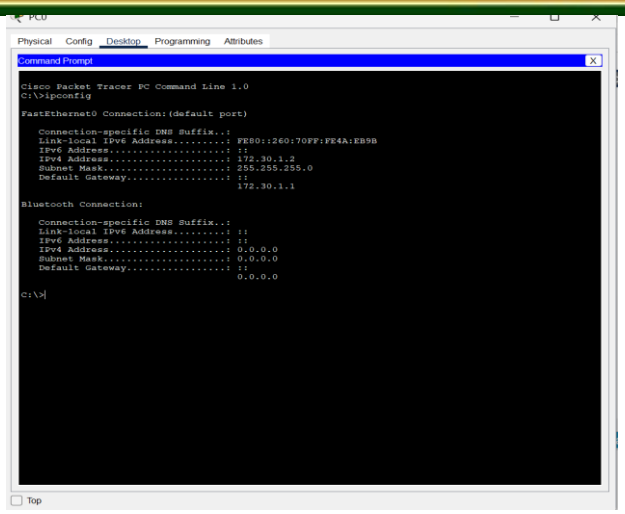


Figure 2.1: IPv4 Configuration for PC0.

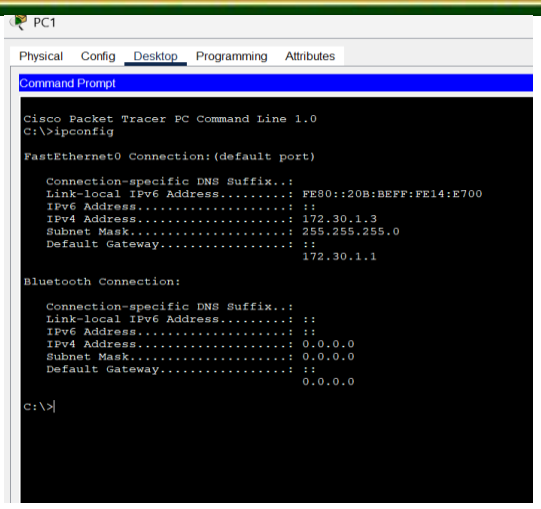


Figure 2.2: IPv4 Configuration for PC1.

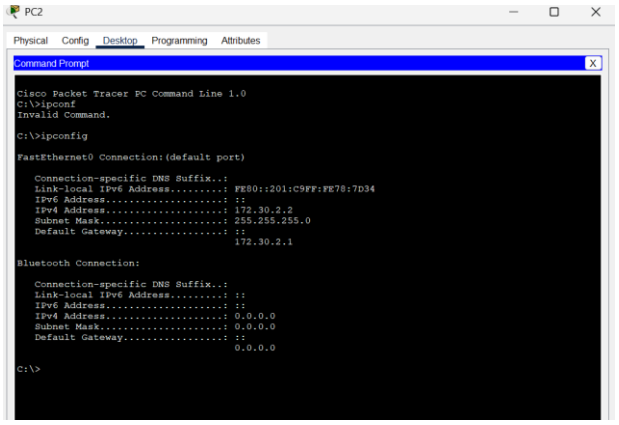


Figure 2.3: IPv4 Configuration for PC2.

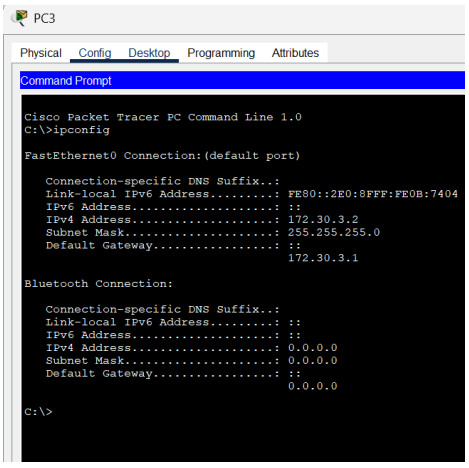


Figure 2.4: IPv4 Configuration for PC3.

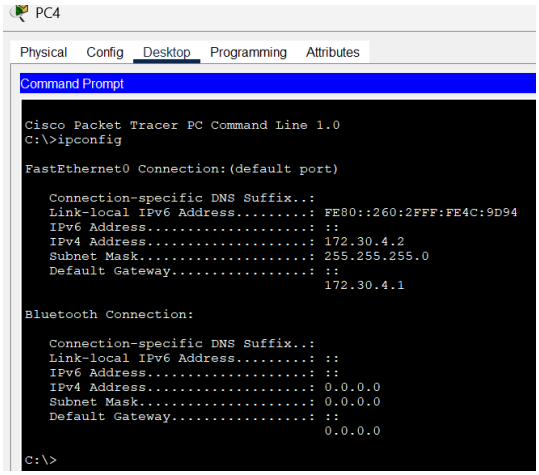


Figure 2.5: IPv4 Configuration for PC4.

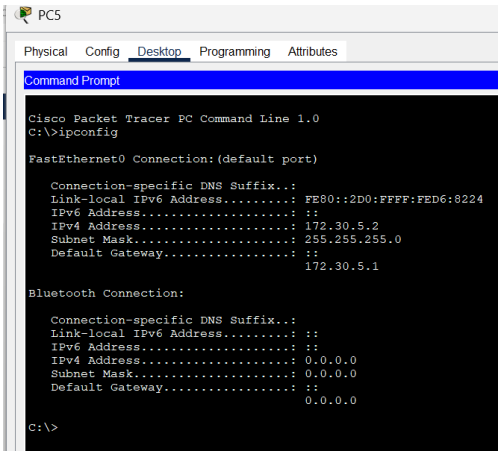


Figure 2.6: IPv4 Configuration for PC5.

3. **Test the ping command between devices in the same Network.**
To verify local connectivity, the ping command was executed between devices within the same subnet. The successful ICMP replies confirm that the Layer 2 switches and Layer 3 IP configurations are functioning correctly for internal communication.

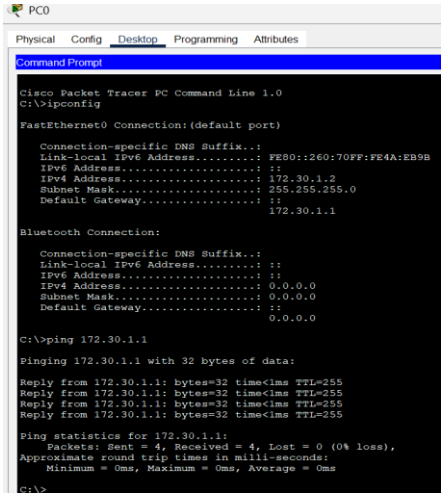


Figure 3.1: Successful Ping from PC0.

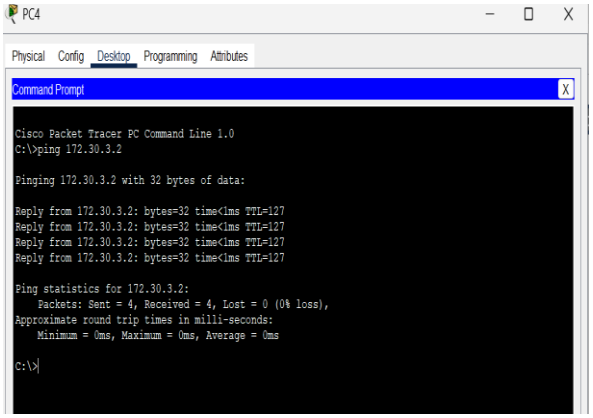


Figure 3.2: Successful Ping from PC4.

4. Add the routing protocol RIP

To enable communication between different networks, the RIP (Routing Information Protocol) was configured on all routers. Using the network command, each router was programmed to advertise its directly connected subnets, allowing the routing table to populate dynamically across the entire topology.



Figure 4.1: RIPv2 Configuration on Router R1.

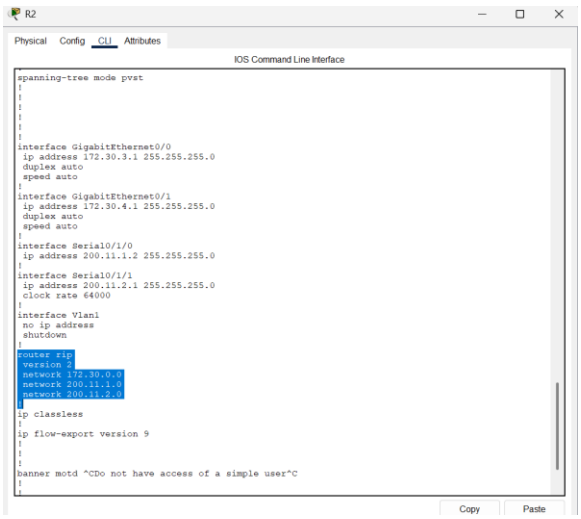


Figure 4.2: RIPv2 Configuration on Router R2.

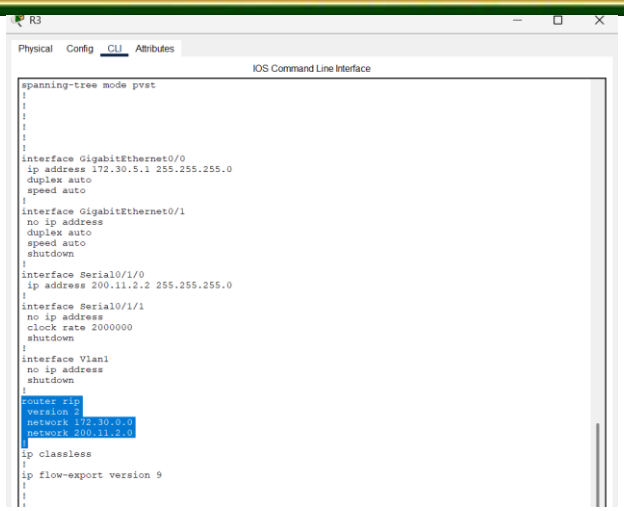


Figure 4.3: RIPv2 Configuration on Router R3.

5. Test the ping command between device in different Network.

After configuring the RIP protocol, a final end-to-end connectivity test was performed. The successful ICMP replies between devices in different networks (e.g., from PC0 to PC4) demonstrate that the routers are correctly exchanging routing updates and that packets are being successfully routed across the entire topology.

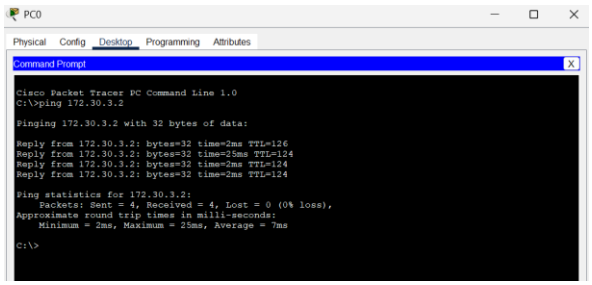


Figure 5.1: Successful Ping from PC0 to a remote network device.

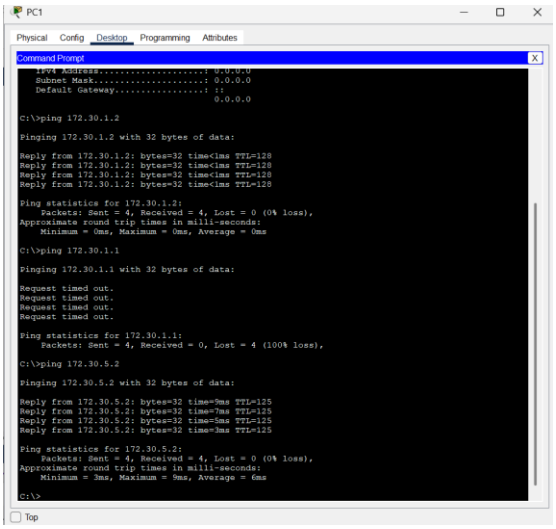


Figure 5.2: Connectivity verification from PC1 across the topology.

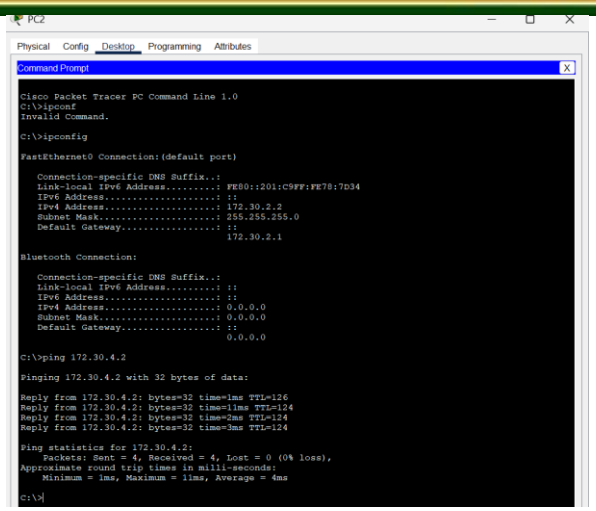


Figure 5.3: Successful communication test from PC2.

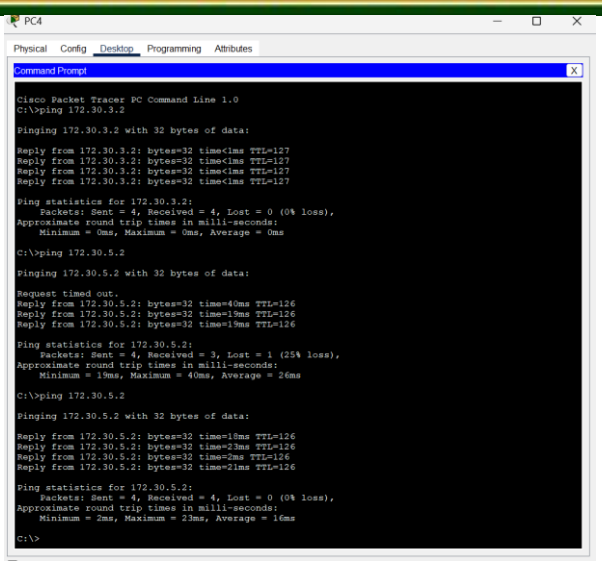


Figure 5.4: Final connectivity verification from PC4.